Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of AI-Based Design and Manufacturing Process Technologies for Alumina-Based Ballistic Materials Used in Wheeled Armored Vehicles**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		***Mechanical・Materials***	
	Medium Category	Small Category	Medium Category	Small Category
	Ceramic Materials	***Ceramic Manufacturing Process Technologies***	***Plastic deformation / powder***	***Powder processing technologies***
Concept of Proposed Technology	**Core Concept** Conventional alumina (Al2O3) faces a performance ceiling due to the trade-off between hardness and fracture toughness. The current trial-and-error development process is too costly and slow to validate complex variables or counter rapidly evolving ballistic threats. To overcome this technological bottleneck, we propose shifting from experiment-heavy methods to an AI and simulation-driven paradigm. This approach will drastically shorten R&D cycles and surpass existing material limits. The technology proposed in this project is a "Closed-Loop AI-Driven Ballistic Material Design System." It is an intelligent material development platform that integrates Korean SME partner's extensive manufacturing and evaluation data with Purdue University's advanced AI-based inverse design and simulation expertise.			

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Rationale for Joint R&D Project

Key Functions and Potential Applications

- Closed-Loop System: A recursive R&D framework where physical experimental data trains the AI models (Virtual), and the optimal process conditions derived by the AI are, in turn, validated through subsequent physical experiments.
- Inverse Design: Upon inputting target ballistic performance requirements (Target Properties), the AI inversely calculates the optimal raw material mixing ratios and the Sintering process schedules required to achieve them.
- Innovation: Moving beyond simple process refinement, this platform transforms the decades-long reliance on experience-based development into "Data-Driven Engineering." This paradigm shift will enable the realization of "Super-Gap" alumina ballistic ceramics that possess both the cost-competitiveness to replace high-priced materials and world-class protection performance.

Critical Technical Barriers

1. Technical Challenges in Advanced Bulletproof Ceramic Development

The most critical technical challenge in developing ballistic ceramics, such as alumina (Al2O3), lies in overcoming the trade-off between hardness and fracture toughness. Generally, increasing hardness reduces toughness, leading to brittle fracture upon impact; conversely, enhancing toughness often compromises hardness, thereby reducing penetration resistance.

To simultaneously improve both properties, advanced techniques are employed, such as introducing reinforcing agent as a reinforcing phase, adding various metal oxide dopants (e.g., and implementing complex processes like process-controlled sintering. However, these multi-component and multi-stage process variables are highly coupled. Identifying the "Sweet Spot"—the optimal process conditions that simultaneously achieve grain growth inhibition and full densification—is an extremely sophisticated problem, requiring the exploration of a mathematically vast and complex design space.

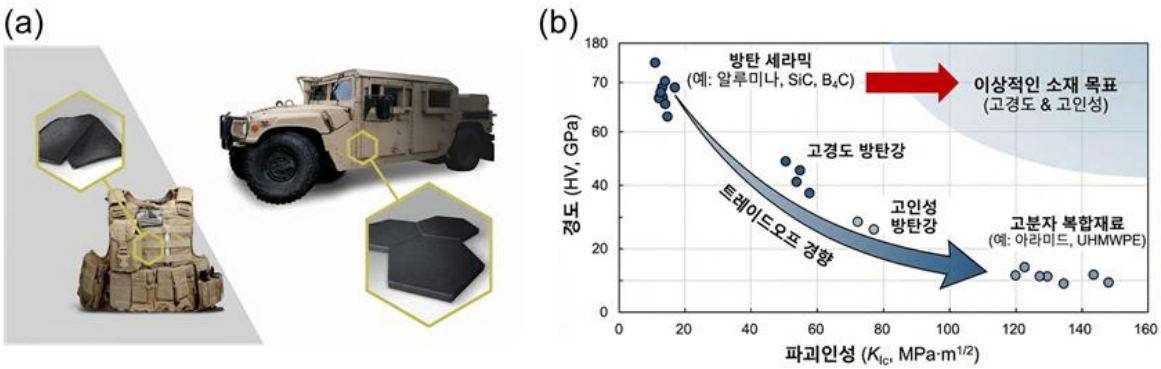

Figure 1. (a) Ballistic protection material for personal and wheeled armored vehicles. (b) Simultaneous Enhancement of Hardness and Toughness in Ballistic Materials [1-3]

2. Limitations of Conventional Trial-and-Error Methodologies

Current ballistic material development relies on trial-and-error, where specimens are fabricated based on intuition and validated through destructive testing. However, ballistic testing is cost-prohibitive and time-intensive, making it practically impossible to experimentally verify the vast array of process variables. Furthermore, the lack of clarity in identifying whether performance failures stem

from material composition or sintering processes limits effective feedback. This inefficiency creates a critical bottleneck, hindering the time-to-market delivery of high-performance materials essential to countering rapidly evolving adversarial threats.

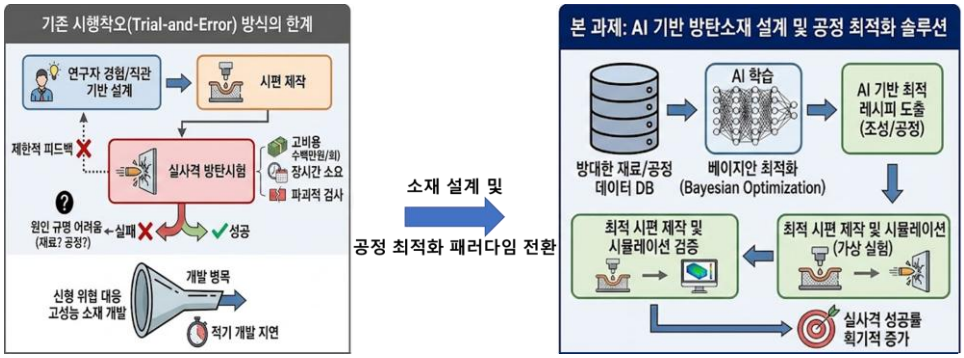

Figure 2. Limitations of trial-and-error methods and our proposed AI-driven solution

Why Global Collaboration?

To lead the next-generation ballistic materials market, strategic collaboration between manufacturing leaders and virtual design experts is essential. Purdue University possesses world-class expertise in AI-based inverse design and Bayesian optimization. Utilizing the JH-2 (Johnson-Holmquist) constitutive model and high-speed impact simulations, Purdue can precisely predict protection performance based on microstructure data without physical firing tests. This joint R&D integrates Korean SME partner's manufacturing data with Purdue's AI/simulation technologies to shift the development paradigm toward Data-Driven Engineering. The goal is to drastically reduce R&D costs and cycles, securing independent design capabilities to achieve a "Super-Gap" technical advantage in the global defense market.

Key Words	1. Alumina-based ceramic 2. Bulletproof ceramic 3. AI-based inverse design 4. Sintering 5. High-speed collision simulation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

Overall Objective

The objective is to establish an "Intelligent Ballistic Material Design and Manufacturing Platform" by integrating Purdue's AI inverse design with Samyang's manufacturing expertise. By shifting to Data-Driven Engineering, we aim to overcome the hardness-toughness trade-off, shorten development cycles by over 50%, and secure "Super-Gap" technologies to maximize the survivability of wheeled armored vehicles.

Detailed Objectives

- Specific Objective 1: AI-Based Inverse Design & Impact Prediction
  We will develop an "Inverse Design Algorithm" that derives optimal material compositions and sintering conditions for target performance. Using Bayesian Optimization and JH-2 constitutive models, Purdue will establish a Digital Twin system to precisely predict ballistic performance, minimizing the need for physical firing tests.
- Specific Objective 2: Controlled Sintering and Microstructure Control
  This goal focuses on advancing controlled Sintering to simultaneously achieve grain growth inhibition and densification. The Korean SME partner will fabricate specimens (e.g., ZTA, doped alumina) and provide analysis data to the AI Feedback Loop, identifying physical correlations to establish standardized, mass-production-ready processes.
- Specific Objective 3: Performance Validation & Design Guidelines
  We will ensure model reliability by validating AI predictions against Ground Truth ballistic data. The final output will be "Ballistic Ceramic Design Guidelines for Wheeled Armored Vehicles," providing customized recipes for evolving threats and establishing a technical foundation for global market competitiveness.

R&D Target Specifications

Parameters	Goal
Hardness	≥ **17 GPa**
Fracture toughness	≥ **4.0 MPaOm$^{1/2}$**
Relative density	≥ **99 %**
Elastic modulus	≥ **350 GPa**
Flexural strength	≥ **350 MPa**
Protection level	**STANAG 4569 Lv.4** (≤ 95% of Current Standard Areal Density)
AI model Prediction accuracy	≥ **90 %** (vs. Ground truth)

Table 1. Quantitative target specifications

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R&D Plan	● Year 1 (6 months): Data Framework & Microstructure Modeling Korean SME partner ■ Standardization of alumina sintering & baseline data acquisition ■ Fabrication & characterization of single-doped sintered bodies ■ Data extraction and feedback for AI model optimization Purdue University ■ Microstructure modeling ■ AI training with microstructural data ■ Initial AI surrogate model ● Year 2 (12 months): Virtual Process Optimization & Active Learning Korean SME partner ■ Multi-component composition & microstructure control ■ Optimization for grain growth & residual pore minimization ■ Heating rate & thermal shock prevention during the sintering process ■ Validation of AI-recommended compositions Purdue University ■ Density functional theory (DFT) calculation under dopant implementation ■ Optimization modeling of the sintering process ■ Bayesian optimization-based composition discovery ● Year 3 (12 months): Ballistic Performance Prediction & System Verification Korean SME partner ■ Phase control & multi-additive interaction ■ Sintering optimization for pore elimination ■ Ballistic performance evaluation & STANAG 4569 Validation Purdue University ■ High-speed impact simulation via JH-2 (Johnson-Holmquist-2) constitutive model ■ Completion of AI-Based inverse design algorithm ● Year 4 (6 months): Platform Finalization & Tech Transfer Korean SME partner ■ Optimization schedule for grain & pore control ■ AI-Based inverse design system implementation Purdue University ■ Integrated process-property-performance pipeline for Digital Twin ■ Digital Twin Consistency & Automated Calibration

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● AI Modeling & Data analytics ■ Microstructure modeling ■ AI training with microstructural data ■ Initial AI surrogate model ● Virtual Design & Bayesian Optimization ■ Density functional theory (DFT) calculation under dopant implementation ■ Optimization modeling of the sintering process ■ Bayesian optimization-based composition discovery ● Multi-scale Microstructure Simulation & High-Fidelity Ballistic Impact Analysis ■ High-speed impact simulation via JH-2 (Johnson-Holmquist-2) constitutive model ■ Completion of AI-Based inverse design algorithm ■ Optimization schedule for grain & pore control ■ AI-Based inverse design system implementation
Expected R&R of Korean SME Partner	● Material Mixing & Processing ■ Standardization of alumina sintering & baseline data acquisition ■ Fabrication & characterization of single-doped sintered bodies ■ Data extraction and feedback for AI model optimization ● Optimization Implementation ■ Multi-component composition & microstructure control ■ Optimization for grain growth & residual pore minimization ■ Heating rate & thermal shock prevention during the sintering process ■ Validation of AI-recommended compositions ● Testing & Validation ■ Phase control & multi-additive interaction ■ Sintering optimization for pore elimination ■ Ballistic performance evaluation & STANAG 4569 Validation ● Ground Truth Data Provision ■ Optimization schedule for grain & pore control ■ AI-Based inverse design system implementation

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Economic impact ● Cost competitiveness and efficiency: reduce development cycles by over 50% and lower defect rates to under 10% through ai inverse design and sintering optimization, securing manufacturing cost advantages. 2. Export performance ● Global supply chain entry: enhance the global competitiveness of mobility equipment equipped with ai-designed high-performance, lightweight ballistic ceramics. leverage international partnerships established through k2 tank exports to Poland and Italy tank technology transfers to enter global defense supply chains. 3. Enhancement of weapon systems ● Improved mobility and protection: reduce armor weight by over 10% while maintaining protection levels using ai-derived high-performance alumina. improve acceleration, climbing ability, and fuel efficiency of wheeled armored vehicles and main battle tanks for enhanced long-range deployment. 4. Advancement of science and technology ● Paradigm shift in materials R&D: present a next-generation research direction of "data-driven materials development" by integrating ai inverse design with traditional ceramic processes, shifting from trial-and-error to digitalized property prediction.

6. References

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[3] A. Bhatnagar, “Introduction to ballistic protection materials”, in Lightweight ballistic composites: Military and law-enforcement applications (2nd ed.), Woodhead Publishing (2016)